

32T CSCA – 2022
Computer Science and Applications (Theory) – Solved

Q1. Answer the following (any eight): [1×8=8]

(a) State two advantages of C++.

Ans: C++ supports Object Oriented Programming (data security through encapsulation, code reuse through inheritance), and it is a fast, efficient language that gives low-level access to memory (through pointers) while also supporting high-level abstractions.

(b) What is a class?

Ans: A class is a user-defined data type that acts as a blueprint for creating objects. It groups data members (variables) and member functions (methods) that operate on that data into a single unit.

(c) What is an object?

Ans: An object is an instance of a class. It is a real-world entity created using a class definition, having its own copy of data members and access to the class's member functions.

(d) State two data types of C++.

Ans: int (used to store integer values) and float (used to store decimal/real numbers) are two basic data types of C++.

(e) Define a data structure.

Ans: A data structure is a particular way of organizing, storing and managing data in a computer so that it can be accessed and modified efficiently, e.g. array, stack, queue, linked list.

(f) What is member function?

Ans: A member function is a function that is declared and defined inside a class and operates on the data members of that class. It defines the behaviour/operations that objects of the class can perform.

(g) What is a database?

Ans: A database is an organized collection of related data stored electronically, which can be easily accessed, managed, and updated using a Database Management System (DBMS).

(h) What is SQL?

Ans: SQL (Structured Query Language) is a standard language used to create, access, manipulate and manage data stored in a relational database, using commands like SELECT, INSERT, UPDATE and DELETE.

(i) What is a binary file?

Ans: A binary file is a file that stores data in binary (0s and 1s) format exactly as it is held in memory, rather than as readable text. It is compact and faster to read/write but not human-readable.

(j) Define constructor.

Ans: A constructor is a special member function of a class that has the same name as the class, has no return type, and is automatically called when an object of that class is created, mainly used to initialize data members.

Q2. Answer the following questions (any ten): [2×10=20]

(a) What is function overloading?

Ans: Function overloading is a feature of C++ where two or more functions can have the same name but differ in the number or type of their parameters. The compiler selects the correct function based on the arguments passed during the function call.

(b) Define abstract class and concrete class.

Ans: An abstract class is a class that contains at least one pure virtual function and cannot be instantiated directly; it is meant to be used as a base class. A concrete class is a class in which all member functions are fully defined, and objects can be created from it directly.

(c) Define a destructor.

Ans: A destructor is a special member function with the same name as the class preceded by a tilde (~), having no return type and no arguments. It is automatically invoked when an object goes out of scope or is deleted, and it is used to release resources allocated to the object.

(d) State the commutative and associative law of Boolean algebra.

Ans: Commutative law: $A + B = B + A$ and $A \cdot B = B \cdot A$. Associative law: $(A + B) + C = A + (B + C)$ and $(A \cdot B) \cdot C = A \cdot (B \cdot C)$.

(e) What is packet switching?

Ans: Packet switching is a method of data transmission in which data is broken into small units called packets, each sent independently through the network (possibly via different routes) and reassembled at the destination.

(f) Define Bandwidth.

Ans: Bandwidth is the maximum amount of data that can be transmitted over a network or communication channel in a given amount of time, usually measured in bits per second (bps).

(g) State two uses of XML.

Ans: XML is used to store and transport data between different systems/applications, and it is used to create custom, self-descriptive tags for structuring data in a platform-independent way (e.g. in web services and configuration files).

(h) Define a Queue.

Ans: A queue is a linear data structure that follows the FIFO (First In First Out) principle, where insertion of elements (enqueue) takes place at the rear end and deletion (dequeue) takes place at the front end.

(i) Define a Stack.

Ans: A stack is a linear data structure that follows the LIFO (Last In First Out) principle, where insertion and deletion of elements both take place from the same end, called the top of the stack.

(j) State two advantages of LAN.

Ans: A LAN allows easy sharing of resources such as printers and files among connected computers, and it provides high-speed, low-cost data communication within a small geographical area such as an office or building.

(k) What is data transfer rate?

Ans: Data transfer rate is the speed at which data is transmitted from one device/point to another over a network, usually measured in bits per second (bps), kbps, or Mbps.

(l) State the use of Karnaugh Map.

Ans: A Karnaugh Map (K-Map) is a graphical method used to simplify Boolean algebraic expressions/logic functions, reducing the number of gates required to implement a digital circuit.

(m) What is a base class?

Ans: A base class (or parent class) is the class from whose properties and members another class (derived/child class) inherits, in the concept of inheritance in object-oriented programming.

(n) Define Pointer.

Ans: A pointer is a variable that stores the memory address of another variable, rather than storing a value directly. It is declared using the * symbol, e.g. `int *p;`

Q3. Answer any nine from the following: [3×9=27]

(a) Write briefly about the advantages of object oriented programming language.

Ans: OOP languages provide data security through encapsulation (data and functions bound together, hidden from outside access), code reusability through inheritance (a new class can reuse the properties of an existing class), and flexibility through polymorphism (the same function/operator behaves differently in different contexts). They also make large software easier to design, maintain and debug by modelling real-world entities as objects.

(b) State the process of creating and writing text into a file.

Ans: In C++, an output file stream object is created using the ofstream class. The file is opened either through the constructor or the open() function, e.g. ofstream fout("data.txt"); Text can then be written into the file using the insertion operator (<<), e.g. fout << "Hello"; After writing, the file must be closed using the close() function, e.g. fout.close(), to save the data properly.

(c) Differentiate 'pass by value' from 'pass by reference'.

Ans: In pass by value, a copy of the actual argument's value is passed to the function, so changes made inside the function do not affect the original variable. In pass by reference, the address (reference) of the actual argument is passed, so changes made inside the function directly affect the original variable outside the function.

(d) Write the codes in C++ to declare and initialize a pointer.

```
int a = 10;
int *p;      // declaring a pointer
p = &a;      // initializing pointer with address of a
```

Here, p is a pointer that stores the address of variable a.

(e) Write SQL to create a table named student with 4 fields namely: SID, SNAME, ADDRESS and MARKS.

```
CREATE TABLE student (
    SID INT,
    SNAME VARCHAR(30),
    ADDRESS VARCHAR(50),
    MARKS INT
);
```

(f) Write the differences of Text file and Binary file.

Ans: A text file stores data in human-readable ASCII/character format, whereas a binary file stores data in the same binary form as it is held in memory. A text file is generally larger in size and slower to access, whereas a binary file is more compact and faster to read/write. Text files can be opened and edited using a normal text editor, but binary files require a specific program to interpret their contents correctly.

(g) What are the advantages of using SQL?

Ans: SQL is easy to learn and use with simple English-like commands. It allows fast retrieval, insertion, updating and deletion of large volumes of data. It is portable and works across many different database systems, and it also allows multiple users to access the database simultaneously while maintaining data integrity and security.

(h) Explain the Push and POP operations on a stack.

Ans: Push is the operation of inserting a new element onto the top of the stack; before pushing, the program checks for a stack overflow condition (when the stack is full). Pop is the operation of removing/deleting the element present at the top of the stack; before popping, the program checks for a stack underflow condition (when the stack is empty). Both operations affect only the top of the stack, following the LIFO principle.

(i) Explain the process of selection sort with an example.

Ans: In selection sort, the smallest element from the unsorted part of the list is repeatedly selected and swapped with the element at the beginning of the unsorted part. This process is repeated for the remaining unsorted portion until the whole list is sorted.

Example: Consider the list 5, 2, 4, 1

Pass 1: smallest is 1 -> swap with 5 => 1, 2, 4, 5

Pass 2: smallest in remaining (2,4,5) is 2 -> already in place => 1, 2, 4, 5

Pass 3: smallest in remaining (4,5) is 4 -> already in place => 1, 2, 4, 5
Sorted list: 1, 2, 4, 5

(j) Explain the process of obtaining Sum of Product (SOP) using Truth Table.

Ans: To obtain the SOP form from a truth table, we look at all the rows where the output is 1. For each such row, a product (AND) term is written by taking each input variable in its true form if it is 1, and in its complemented form if it is 0. Finally, all these product terms are combined using the OR (+) operation to get the Sum of Products expression, e.g. for a function with output 1 at A=0,B=1 and A=1,B=1, the SOP would be $F = A'B + AB$.

(k) Explain the aggregate functions of SQL.

Ans: Aggregate functions in SQL perform a calculation on a set of values and return a single summary value. Common aggregate functions are: COUNT() – counts the number of rows/values, SUM() – adds up numeric values, AVG() – finds the average value, MAX() – finds the maximum value, and MIN() – finds the minimum value in a specified column.

(l) Explain three different types of network topologies.

Ans: Bus topology: all devices are connected to a single central cable (the bus); it is simple and low-cost but a fault in the main cable disrupts the whole network. Star topology: all devices are connected individually to a central hub/switch; it is easy to manage and a single node failure does not affect others, but if the hub fails the whole network is affected. Ring topology: each device is connected to exactly two other devices, forming a closed loop, and data travels in one direction around the ring; it is orderly but a break in any link can disrupt communication.

(m) Differentiate LAN from WAN.

Ans: A LAN (Local Area Network) covers a small geographical area such as a single building or campus, offers high data transfer speed, and is relatively inexpensive to set up. A WAN (Wide Area Network) covers a large geographical area such as a country or the whole world, generally has a comparatively lower data transfer speed, and is more expensive to set up, often using leased telecommunication lines or satellites.

(n) Write the advantages of Java Script.

Ans: JavaScript is a lightweight, client-side scripting language that runs directly in the browser, reducing server load and providing faster response since it does not need to communicate with the server for simple tasks. It allows creation of interactive and dynamic web pages (e.g. form validation, animations) and is supported by virtually all modern web browsers without requiring any extra plugin.

Q4. Answer any three from the following: [5×3=15]

(a) Write a C++ program to add two matrices of order 4×4 .

```
#include <iostream>
using namespace std;
int main() {
    int a[4][4], b[4][4], c[4][4];
    cout << "Enter elements of first matrix:\n";
    for (int i = 0; i < 4; i++)
        for (int j = 0; j < 4; j++)
            cin >> a[i][j];

    cout << "Enter elements of second matrix:\n";
    for (int i = 0; i < 4; i++)
        for (int j = 0; j < 4; j++)
            cin >> b[i][j];

    for (int i = 0; i < 4; i++)
        for (int j = 0; j < 4; j++)
            c[i][j] = a[i][j] + b[i][j];
```

```

        cout << "Sum of the two matrices:\n";
        for (int i = 0; i < 4; i++) {
            for (int j = 0; j < 4; j++)
                cout << c[i][j] << " ";
            cout << endl;
        }
        return 0;
}

```

(b) Write a C++ program to implement binary search in an array.

```

#include <iostream>
using namespace std;
int main() {
    int n, key, low, high, mid;
    cout << "Enter number of elements: ";
    cin >> n;
    int arr[n];
    cout << "Enter elements in ascending order:\n";
    for (int i = 0; i < n; i++)
        cin >> arr[i];

    cout << "Enter the element to search: ";
    cin >> key;

    low = 0; high = n - 1;
    bool found = false;
    while (low <= high) {
        mid = (low + high) / 2;
        if (arr[mid] == key) {
            cout << "Element found at position " << mid + 1 << endl;
            found = true;
            break;
        }
        else if (arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    if (!found)
        cout << "Element not found" << endl;
    return 0;
}

```

(c) Explain the concept of multiple inheritance.

Ans: Multiple inheritance is a feature of object-oriented programming in which a single derived class is created from more than one base class, thereby inheriting the properties and behaviours of all the base classes. For example, a class 'Result' may inherit from two base classes 'Test' and 'Sports', combining the features of both. Its general syntax in C++ is:

```

class Result : public Test, public Sports {
    // body of derived class
};

```

Multiple inheritance allows greater code reuse, but it can also lead to ambiguity if the base classes have members with the same name (this is usually resolved using virtual base classes).

(d) Compare XML with HTML.

Ans: HTML (HyperText Markup Language) is designed to display data and focuses on how the data looks, using a fixed set of predefined tags (like <p>, <table>,). XML (eXtensible Markup Language) is designed to store, carry and transport data and focuses on what the data is, allowing users to define their own custom tags. HTML is not case-sensitive and is somewhat tolerant of unclosed tags, whereas XML is case-sensitive and strictly requires every tag to be properly closed.

(e) Write briefly about DDL and DML with example.

Ans: DDL (Data Definition Language) consists of SQL commands used to define, create, alter or delete the structure of database objects such as tables. Examples include CREATE, ALTER and DROP, e.g. CREATE TABLE student(SID INT, SNAME VARCHAR(20)); DML (Data Manipulation Language) consists of SQL commands used to manipulate/manage the actual data stored within those tables. Examples include INSERT, UPDATE, DELETE and SELECT, e.g. INSERT INTO student VALUES (1, 'Rahul');

(f) Write a C++ program to delete an element from an array.

```
#include <iostream>
using namespace std;
int main() {
    int n, pos;
    cout << "Enter number of elements: ";
    cin >> n;
    int arr[n];
    cout << "Enter the elements:\n";
    for (int i = 0; i < n; i++)
        cin >> arr[i];

    cout << "Enter position to delete (1 to " << n << "): ";
    cin >> pos;

    for (int i = pos - 1; i < n - 1; i++)
        arr[i] = arr[i + 1];
    n--;

    cout << "Array after deletion:\n";
    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";
    cout << endl;
    return 0;
}
```